Yu-Han Lee (Tyler)

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Summary

MSc Advanced Computing student at King's College London with experience in system software, AI infrastructure, and computer vision. Skilled in Python, C++, PyTorch, Docker, GCP, and HPC environments, with hands-on experience building modular ML pipelines and scalable systems. Published research (TAAI 2023) and delivered high-performance video analytics solutions, achieving 97% detection accuracy and 86% action recognition.

Work Experience

Chung Yuan Christian University — Taoyuan, Taiwan
Undergraduate Research Assistant (NSTC Funded) — Jul 2023 – Feb 2024

- Conducted research on **one-shot learning for face recognition** in classroom roll call systems.
- Implemented deep learning pipelines with **Python and PyTorch (Triplet Network, Transfer learning)**.
- Awarded **NSTC Undergraduate Research Grant** (one of <5% accepted projects nationwide).
- Co-authored **TAAI 2023** paper presenting the system and experimental results.

Chang Ching Yu Memorial Library, Chung Yuan Christian University — Taoyuan, Taiwan
Library IT Intern — Feb 2022 – Feb 2024

- Enhanced a Python-based thesis checker, reducing processing time by **25%** and improving **accuracy by 30%**.
- Provided technical support to international students, facilitating communication during the review process.
- Automated workflow scripts (e.g., BibTeX conversion), improving **reliability by 20%** and **reducing resolution time by 15%**.

Taiwan Mobile Co., Ltd. — Taipei, Taiwan
Software Engineer Intern — Jul 2023 – Aug 2023

- Optimized website performance, improving **SEO rankings by 20%**, increasing traffic by 15%.
- Formulated Java-based image conversion tools, boosting processing speed and accuracy.
- Devised methods for activating mobile cameras using HTML5, W3C MediaStream API, Vue.js, and jQuery.
- Designed a YOLOv8-based web scraper to identify and replace company logos across multiple websites.

Projects

MSc Thesis Project – Automated Basketball Video Analysis — Feb 2025 – Jul 2025

- Designed a modular CV pipeline integrating YOLOv8 (detection), ByteTrack (tracking), R(2+1)D (action recognition), CLIP (team classification), and homography (tactical visualization).
- Deployed on HPC cluster with PyTorch and Linux; achieved YOLOv8 mAP 0.97 and R(2+1)D 86% accuracy.
- Delivered tactical insights such as player trajectories, speed, and team strategies.

BSc Graduation Project – One-shot Learning for Classroom Roll Call — Sep 2022 – Sep 2023

- Researched Few-shot learning using deep learning to address face recognition challenges in a student roll call system.
- Combined data augmentation, transfer learning, and Triplet Network to train a model with one photo per student, achieving accurate face recognition.

– Proposed and verified a research plan for a fellowship from the National Science and Technology Council, Taiwan (acceptance rate <5%).

– Published paper at TAAI 2023 (Domestic Track).

Our-Scheme (Programming Language Project) Feb 2023 – Jul 2023

– Built a custom C++ interpreter for processing and evaluating expressions based on formal language specifications.

– Engineered lexical analysis and syntax parsing modules to tokenize and interpret user input.

5-stage Pipelined CPU (Computer Organization Project) Feb 2022 – Jul 2022

– Designed and implemented a 5-stage pipelined MIPS-Lite CPU using Verilog HDL and ModelSim.

– Supported arithmetic, logic, memory, and branching operations; validated through waveform simulations.

SIC/SICXE Assembler (System Programming Project) Aug 2021 – Jan 2022

– Developed a two-pass assembler in C++ for SIC and SIC/XE machine languages.

– Translated source code into object code with syntax validation and instruction set support.

Education

King's College London London, United Kingdom
MSc Advanced Computing Sep 2024 – Sep 2025

– **Relevant Coursework:** Artificial Intelligence Planning, Multi-Agent Systems, Computer Vision, Security Engineering, Big Data Technologies, Distributed Ledgers & Crypto-Currencies, Pattern Recognition & Deep Learning, Software-Engineering and Underlying Technology for Financial Systems.

– **Thesis:** *"Towards Automated Basketball Video Analysis through Modular Deep Learning Architectures: Detection, Recognition, and Tactical Insight"*.

– Built a modular CV system (YOLOv8, ByteTrack, R(2+1)D, CLIP, homography) and deployed on HPC cluster with PyTorch and Linux.

Chung Yuan Christian University Taoyuan, Taiwan
BSc Information and Computer Engineering Sep 2020 – Jan 2024
Department of Commercial Design (Product Design Division) Sep 2019 – Sep 2020

– **GPA:** 3.93/4.0 (Dean's List, Ranked 2^{nd}/56 (Top 3%) in Spring 2020).

– **Overall:** 88.79%

– **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Organization, System Programming, Computer Networks, Probability & Statistics, Data Science & Applications.

Publications

Lin, W.-J., **Y.-H. Lee**, W.-H. Hsu, and C.-C. Yu (2023). "One-shot Learning for Classroom Roll Call System using Triplet Network and Transfer Learning". In: *Proceedings of the 28th Conference on Technologies and Applications of Artificial Intelligence (TAAI 2023)*. Domestic Track, Dec. 1–2, 2023, National Yunlin University of Science and Technology, Yunlin, Taiwan.

Skills

Languages	Python, C++, Java, JavaScript
ML / CV	PyTorch, YOLOv8, R(2+1)D, CLIP, OpenCV, TensorFlow, Keras
Systems / Infra	Docker, Linux, Git, HPC (High Performance Computing), GCP (Google Cloud Platform)
Data / Tools	NumPy, pandas, Matplotlib, SQL
Other	Verilog, ImageMagick